Heart Disease Classification with Regularized Logistic Regression

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***Abstract*—We classify self-reported coronary heart disease from BRFSS survey data using L2-regularized logistic regression implemented in NumPy. Our pipeline removes highly incomplete columns, imputes remaining missing values, and standardizes predictors. After excluding one malformed record, 328,134 labeled samples remain, of which 8.83% are positive. A five-fold search selects a regularization strength of 1 and a step size of 0.05. On a separate stratified validation set of 65,627 samples, the model achieves F1 0.3773 and accuracy 0.8323. Error analysis reveals moderate positive-class recall but substantial false-positive rates.**

I. Introduction

We predict the MICHD label from demographic, lifestyle, and health-related survey variables. Labels $+1$ and -1 indicate reported heart disease and its absence, respectively. This task concerns an observed survey outcome, rather than a validated forecast of future disease onset. The supplied training table contains 328,135 samples and 322 columns, including an identifier; excluding it leaves 321 candidate predictors.

Class imbalance motivates reporting positive-class F1 alongside accuracy. Our study evaluates a single model family, regularized logistic regression, and examines sensitivity to its regularization strength and optimization step size. No separately trained baseline or controlled preprocessing ablation was evaluated. All learning computations use NumPy and the Python standard library, with Matplotlib used for visualization.

II. Models and Methods

A. Data preparation

The loader excludes one malformed feature record with 321 fields instead of 322 and removes the label at the corresponding original row position. This leaves 328,134 samples: 299,159 negative and 28,975 positive (8.8302%). The loader assumes the original feature and label files share row order; array lengths alone do not independently establish identifier alignment.

A stratified 80/20 split with seed 42 produces 262,507 training and 65,627 validation samples. Both partitions contain approximately 8.83% positive labels. Preprocessing is fitted on the training partition only. Columns with more than 80% NaNs are removed: 116 of 321 predictors are discarded, leaving 205. Remaining NaNs are replaced by training-column medians. Features are centered and scaled using training means and standard deviations, with zero standard deviations replaced by one. A constant intercept column yields design matrices with 206 columns. The same transformations are applied to validation and test data. Numeric survey response codes are not automatically converted to missing values.

B. Classifier and loss implementation

Labels are mapped to $\{0, 1\}$ for training. With the intercept included in x_i, predictions are $p_i = \sigma(x_i^\top w)$, where $\sigma(z) = (1 + e^{-z})^{-1}$. The intended regularized objective is

$$J(w) = -\frac{1}{N}\sum_{i=1}^{N}[y_i \log p_i + (1-y_i)\log(1-p_i)] + \lambda\|w\|_2^2. \quad (1)$$

Full-batch gradient descent, initialized at zero, uses

$$w \leftarrow w - \gamma\left[\frac{X^\top(p-y)}{N} + 2\lambda w\right]. \quad (2)$$

All weights, including the intercept, are penalized. Classification uses $p_i \geq 0.5$ for label $+1$, and -1 otherwise; no threshold sweep was performed.

The loss helper provided in our implementation evaluates

$$\widehat{L}_l(w) = -\frac{1}{N}\sum_{i=1}^{N}[y_i \log(p_i + \epsilon) + (1-y_i)\log(1-p_i+\epsilon)] - l\|w\|_2^2, \quad (3)$$

with $\epsilon = 10^{-12}$. This diagnostic differs from J when $l \neq 0$: placing the penalty inside the negated sum gives it a negative sign. The training routine calls the helper with its default $l = 0$, returning unpenalized, numerically stabilized cross-entropy; regularization enters the gradient separately. The observed final training loss is 0.6081. This value alone does not establish convergence.

C. Hyperparameter selection

Five-fold CV on the 80% training partition searches $\lambda \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1\}$ and $\gamma \in \{0.01, 0.05, 0.1, 0.3\}$. Each candidate receives 200 gradient updates; selection maximizes mean fold F1. Folds are shuffled, not stratified, and the two remainder samples are omitted from internal CV. Preprocessing is fitted before internal CV, rather than separately in each fold, which may make inner-fold estimates optimistic. The external validation partition is not used to fit preprocessing or select the grid winner. The selected model is trained for 2,000 updates on the 262,507 training samples, without refitting on the validation partition.

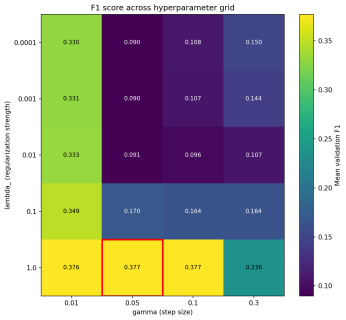


Fig. 1. Mean five-fold validation F1 over regularization strengths and step sizes. The outlined cell is selected; $\gamma = 0.1$ ties its score.

TABLE I
SELECTED MODEL: HELD-OUT VALIDATION METRICS AT THRESHOLD 0.5.

Model	F1	Accuracy	Precision	Recall
L2 logistic	0.3773	0.8323	0.2807	0.5755

III. Results and Analysis

A. Hyperparameter sensitivity

The selected pair is $\lambda = 1$, $\gamma = 0.05$, with mean CV F1 0.3770476. The same mean score is obtained at $\gamma = 0.1$; grid order breaks the tie. At $\gamma = 0.01$, F1 increases from 0.3303 to 0.3762 as regularization increases across the grid. However, larger steps do not consistently improve results: at $\lambda = 1$, increasing γ from 0.1 to 0.3 reduces F1 from 0.3770 to 0.2363 (Figure 1). These differences demonstrate sensitivity to both settings, but do not by themselves diagnose divergence. Because CV and final training use different iteration budgets, the grid reflects both optimization progress and regularization. Fold-level variability was not retained, so uncertainty around mean CV scores is unavailable.

B. Held-out performance

Table I reports performance on 65,627 validation samples. The F1 score of 0.3773 closely matches the selected mean CV score of 0.3770, although one held-out split cannot establish statistical stability. No claim of improvement over another trained model is made.

The confusion matrix contains 51,286 true negatives, 8,546 false positives, 2,460 false negatives, and 3,335 true positives (Figure 2). Thus the model identifies 57.55% of positive cases, misses 42.45%, and incorrectly flags 14.28% of negative cases. Only 28.07% of positive predictions are correct. The predicted positive fraction on validation is 18.10%, above the observed 8.83% prevalence, explaining the substantial false-positive burden. As an arithmetic reference, always predicting negative would yield 91.17% accuracy and zero positive-class F1 on this partition; this is not a separately trained baseline experiment.

C. Coefficient interpretation and test predictions

The largest absolute standardized coefficients include `GENHLTH`, `BPHIGH4`, `EMPLOY1`, and `_AGE80`. Several age-related representations appear among the top coefficients. Their presence is descriptive, not evidence of causal importance: correlated predictors can share effects, and categorical response codes are treated numerically. A negative coefficient for a coded variable cannot be interpreted as a protective clinical effect without checking its coding.

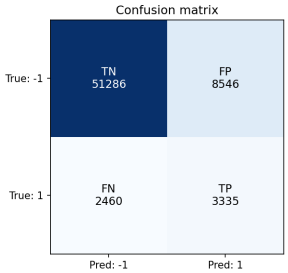


Fig. 2. Validation confusion matrix. Rows are true labels and columns are predicted labels; positive-class recall is 0.5755.

The pipeline generates 109,379 test predictions, with 18.02% classified positive, and writes the original test identifiers with labels to `submission.csv`. No AIcrowd test-set scores were available for this report; all reported performance metrics are local estimates.

IV. Discussion and Conclusion

The current regularized linear classifier achieves validation F1 0.3773 and accuracy 0.8323, recovering over half of positive cases at the cost of many false alarms. The search favors the strongest regularization value tested, but does not establish an optimum beyond this grid. Limitations include unverified identifier alignment, preprocessing outside the inner folds, numerical treatment of categorical codes, possible missing-response sentinels, and the absence of baseline and ablation experiments. Consequently, the individual benefits of imputation, standardization, and regularization cannot be isolated. Future experiments should verify label alignment, fit preprocessing within each fold, assess convergence, and tune the decision threshold using development data only.

Reproducibility

Run `python run.py` with the three supplied CSVs in `dataset/`. Seed 42 controls splits; the grid uses 200 updates per fit and final training uses 2,000. Metrics and figures are written to `reports/`. The submission model uses the 80% training partition. Code: https://github.com/La-ine/ml_cool_kids.